The eigenvalue search in `chladni`: tension-curve measurements, solver requirements, and a triage of candidate methods

Technical report, compiled from `notes/` and `docs/design/`

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Abstract

This report consolidates a measurement session on the eigenvalue-search layer of `chladni`, the method-of-particular-solutions (MPS) solver for Laplacian eigenvalues of planar domains. It has three parts. Part 1 reports the first tension-curve measurements taken on *real* corner domains with corner-adapted Fourier–Bessel bases, rather than on the surrogate or on a single test shape: a seven-domain roster with bases validated against tabulated eigenvalues, the finding that the tension plateau is stable across domains (0.12–0.36) while the attainable floor spreads four to five orders *within* a single scan, an end-to-end recall-versus-stride measurement showing that the project's standing "six samples per gap" rule is wrong by a factor of three, and the identification of the minimum *local* gap — 0.13 to 0.79 of Weyl's mean — as the quantity that actually governs recall. It also records that the Weyl completeness audit reads as a false alarm on every symmetric domain if it is run before multiplicity estimation. Part 2 translates those measurements into a requirements list for the minimization and multiplicity-estimation routines, organized as functional requirements, required diagnostics, and anti-features that were measured *not* to work, each tied to the measurement that forces it. It also records that the harness's search interface must widen from a tension function to a mode function, because the collapse-versus-merge decision needs eigenfunctions. Part 3 reframes the problem as a level-set/certification problem rather than global optimization, enumerates 27 candidate methods, and reports six falsifying tests: Weyl-stratified sampling and the argument principle on $\det A_B$ are dead; pilot-calibrated Lipschitz certification is *unsound*, not merely loose; the exact k-derivative identity holds once a k-dependent column scale is corrected for; DIRECT is strong; and the eigenvector-swing proposal was falsified and reinterpreted into a cell decomposition. The composed candidate (swing cells + DIRECT + refinement) reaches mean recall 0.975 on a five-domain roster at 2.2× the cost of a uniform grid. A prominent self-correction is recorded: an earlier claim that the swing separates pairs 290× tighter than the dip was a statement about *detection*, and had to be repaired with a bisection-toward-the-peak step before it became a statement about *separation*.

Contents

1 Scope, sources, and what is new **4**

1.1 The project-specific machinery, briefly 4

2 The governing principle: what a successful run means **5**
2.1 The rule that decides which constants may be fixed 5
2.2 Assumptions and ground truth 5
2.3 The three clauses 5
2.4 The constants, and not more work than asked 6

Part 1: Tension curves on real corner domains **6**

3 The roster and how the bases were built and validated **6**

4 The plateau is stable across domains; the floor is not, within one scan **7**

5 Recall versus stride: six samples per gap is not enough **8**
5.1 The mechanism: minimum local gap, not Weyl's mean 9

6 The Weyl audit is a post-multiplicity check **9**
6.1 What §20 changes 10

7 The Weyl deficit as a localizer, not a verdict **10**
7.1 The bug it found first, which was the author's 11
7.2 Two structural flaws, both real 11
7.3 Two cautions on the scoring itself 12

Part 2: What the solver has to have **12**

8 Functional requirements **12**

9 Required diagnostics **14**

10 Anti-features: measured not to work **14**

11 The interface has to widen **15**

12 Ranked build order **16**

Part 3: Candidate methods **16**

13 The reframing that generates the list **16**

14 The enumeration **16**
14.1 Family 1 — deterministic sampling, local extraction 16
14.2 Family 2 — bounding and certification 17
14.3 Family 3 — analytic root counting 17
14.4 Family 4 — derivative-based 18
14.5 Family 5 — structure of the branch family 18
14.6 Family 6 — model-based and statistical 18
14.7 Family 7 — exploiting problem structure 19
14.8 Triage 19

15 The six tests **19**
15.1 A2 — Weyl-stratified sampling: falsified 20
15.2 A6 — Lipschitz calibration: unsound at small pilots 20
15.3 A7 — DIRECT, Lipschitz-free: strong 21
15.4 A11 — the argument principle on $\det A_B$: falsified 22
15.5 A14 — the exact k-derivative: confirmed, after a correction 23
15.6 A19 — the eigenvector swing: falsified as proposed, strong as reinterpreted . . 23
15.7 Shortlist after the six tests 24

16 The swing on a pair: a detection result **25**

17 The composed candidate, scored **26**
17.1 The honest ledger 27
17.2 Two efficiency findings 27
17.3 What the roster scoring left untested 28

18 Detection is not separation: a correction to the pair result **28**
18.1 The defect 28
18.2 The repair: a peak is a feature to bisect toward 28
18.3 The remaining frontier 29

19 Open questions and what is untested **29**

Note on two inconsistencies in the sources **30**

1 Scope, sources, and what is new

The layer under discussion sits above `mps`. Its contract is

> given a tension function returning the ascending vector $\sigma_1 \le \sigma_2 \le \cdots$, find every approximate eigenvalue in $[a, b]$ with its multiplicity, using as few evaluations as possible,

and it sees only the tension function and the interval: no domain, no basis, no pencil (search.md §preamble). That restriction is deliberate — it is what makes two candidate searches comparable, and what would let a later finite-element or boundary-integral method reuse the same search. Part 2, §11, records the one measured reason to widen it.

The material reported here is drawn from `notes/landscape_findings.md` §§20–25 (new), `notes/solver_features.md`, `notes/approaches.md`, and `docs/design/search.md` §2. Sections 1–19 of `landscape_findings.md` are earlier context; they are drawn on for background but not re-reported. The supporting scripts are `notes/curve_lab.py` (basis builder, scanner, roster), `notes/curve_study.py` (the three studies of §20), `notes/adaptive_scan.py` (§21), `notes/test_approaches.py` (§22), `notes/swing_pairs.py` (§23), and `tests/searches.py::swing_direct_search` (§§24–25).

1.1 The project-specific machinery, briefly

Three pieces of vocabulary recur and are worth fixing before the measurements.

Tension and the GSVD pencil. MPS assembles, at a trial value λ, a matrix of particular solutions of the Helmholtz equation evaluated at boundary collocation points (A_B) and at interior points (A_I). The *tension* is the generalized singular value spectrum of the pencil (A_B, A_I): it measures how small the boundary residual of a candidate solution can be made while its interior norm is held away from zero. Writing $\sigma_1(\lambda) \le \sigma_2(\lambda) \le \cdots$ for that ascending spectrum, the eigenvalues of the domain appear as local minima of σ_1. The GSVD formulation is used in preference to a determinant or a plain least-squares residual because it is what makes the objective numerically stable; this matters directly to A11 in Part 3. The interior points are a *normalization*, not a quadrature: their only job is to keep $\|A_I c\|$ away from zero (search.md §1).

The cone model. Near an eigenvalue λ_i the measured shape of the first tension is

$$\sigma_1(\lambda) \;\approx\; \sqrt{\varepsilon^2 + c^2(\lambda - \lambda_i)^2}, \tag{1}$$

a hyperbola that looks like a cone with a rounded bottom. It fits the far field to a median relative residual of 1–3% on rect, disk, eq_tri and the L-shape, failing only where the floor approaches the plateau, i.e. where the basis cannot represent the eigenfunction (search.md §1). The two parameters are independent and it matters which is which: ε is the *floor*, σ_1 at the eigenvalue, and is a property of *the basis*; c is the *wall slope*, $d\sigma_1/dk$ away from the eigenvalue, and is a property of *the domain*. On the L-shape at 20, 40 and 60 Fourier–Bessel terms, ε falls $4.0 \times 10^{-5} \to 1.6 \times 10^{-8} \to 1.1 \times 10^{-12}$ while c moves only $1.602 \to 1.595$. The ratio ε/c is the *resolution limit*: the scale in k at which the cone flattens into its quadratic bottom, so two eigenvalues closer than that have overlapping bottoms and cannot be separated. Since c is fixed by geometry, ε alone sets it — which is the concrete form of "the search is bounded by the

basis". Away from all eigenvalues σ_1 sits on a *plateau*, denoted S below; the plateau is what τ_{scan} is measured against.

Multiplicity by (C1) and (C2). Multiplicity μ at a located level is read off the tension spectrum by two conditions: (C1) a *ratio gap* — σ_{j+1}/σ_j is 1.0–1.7 within a multiplet and 10^{11}–10^{15} across the gap to the next branch, so a gap parameter $G = 10$ separates them; and (C2) a *local-minimum* condition — σ_j must itself have a local minimum at that λ, with prominence at least P, on a bracket wide enough to see it. (C2) is load-bearing rather than belt-and-braces: once the separation $d \lesssim \varepsilon/c$, the ratio rule alone reports multiplicity 2 at a *simple* eigenvalue, and (C2) rejects it every time because σ_2 there is a monotone wall (search.md §1), (solver_features §F9).

2 The governing principle: what a successful run means

Every approach below is judged against `docs/design/search.md` §2, the *operational definition*. It is summarized here because nothing later is meaningful without it.

The layer above sets `prec`. The contract is that the user gets eigenvalues to relative accuracy `prec`, or an honest statement that MPS could not reach it. Doing better than asked is always allowed; doing materially more work than `prec` requires is not.

2.1 The rule that decides which constants may be fixed

A constant may be **fixed** only if it is dimensionless and separates two populations whose separation does not shrink as `prec` tightens. Anything carrying units of λ, and anything expressing *how accurate*, must derive from `prec` (search.md §2.1). This rule, rather than any individual number, is what to apply when a new parameter appears — and it is the reason the roster measurement of the plateau in Part 1 is decisive for τ_{scan}.

2.2 Assumptions and ground truth

The pencil — basis, collocation points, regularization *and* rank policy — is given and adequate. The solver may *detect* inadequacy; it is not responsible for repairing it. Adequacy is quantitative: gap $> 20\,\varepsilon/c$ on $[a, b]$, from a direct measurement in which (C1) is correct at $d/(\varepsilon/c) = 23$ and wrong at 0.45 and 0.15 (search.md §2.2).

The ground truth $S(\texttt{prec})$ is the ordered multiset of local minima of σ_1, each repeated by its multiplicity μ_i (the largest μ such that σ_j has a local minimum at λ_i for every $j \le \mu$), with two floating-point scales attached: "has a local minimum" means prominence at least $P = 1 + m\eta$ with a floor $P \ge 2$, and "at λ_i" means two branch minima lie within `tol_merge` = `prec`, relative. S is indexed by `prec` deliberately: two levels 4×10^{-9} apart are *one* level at `prec` $= 10^{-8}$ and *two* at 10^{-12} (search.md §2.3).

2.3 The three clauses

A solver is **successful at** `prec` on $[a, b]$ when all three hold (search.md §2.4).

(1) Completeness and multiplicity.
$\hat{S}$ matches $S(\texttt{prec})$ as a multiset. Among roots reported close together, collapse-versus-merge is decided by the *interior Gram angle*, not by distance: with $u_i = I(\hat{\lambda}_i)\,\mathrm{coef}(\hat{\lambda}_i, 1)$, $|\langle u_1, u_2\rangle| \geq \tau_{\mathrm{gram}}$ means the same eigenfunction (collapse and recount) and $< \tau_{\mathrm{gram}}$ means independent (merge and sum). Distance survives only as a prefilter deciding which pairs are worth testing.

(2) Accuracy.
Each $|\hat{\lambda}_i - \lambda_i|/\lambda_i \leq \texttt{prec}$, *or* the solver reports that $\delta_{\mathrm{res}}(\lambda_i) > \texttt{prec} \cdot \lambda_i$ and returns δ_{res} as the attained bound. There is no predicted-accuracy constant: δ_{res} is measured from the parabola fit the refinement phase already performs.

(3) Honesty.
The reported per-eigenvalue bar contains the true error, and the completeness diagnostics fire whenever (1) fails.

Clause (3) is not decoration. S is unknown on every domain that matters — `L_shape`, `GWW1` and `mushroom` have tabulated prefixes only — so (1) and (2) need a truth-free proxy on a real run. Part 1 is largely about which proxies actually work.

2.4 The constants, and not more work than asked

Fixed (dimensionless, margin grows as `prec` tightens): $P = 1 + m\eta$ with floor 2; $G = 10$; $\tau_{\mathrm{gram}} = 0.5$; the contract $20\,\varepsilon/c$; the Weyl tolerance 0.5; and τ_{scan} *as a fraction of the plateau, never an absolute tension.* Derived from `prec`: `tol_merge` = `prec`, the accuracy target $\texttt{prec} \cdot \lambda$, and the refinement stop $\max(\texttt{prec} \cdot \lambda,\ \delta_{\mathrm{res}})$ (search.md §2.5).

Section 2.6 adds that **completeness is not a precision question** and the scan stride must not tighten with `prec`. That clause is right; the *number* attached to it in the same paragraph — "about six samples per eigenvalue gap" — is what Part 1 falsifies.

Finally, §2.7 records a real ceiling: `tol_merge` = `prec` against pairs resolvable to about 3×10^{-6} relative leaves a factor of 30 at $\texttt{prec} = 10^{-8}$, but below $\texttt{prec} \sim 4 \times 10^{-9}$ it *inverts* — the discretization's own splitting of a degenerate pair ($1.8 \times 10^{-9}\lambda$ for plane waves, $3.9 \times 10^{-9}\lambda$ for Fourier–Bessel) exceeds `tol_merge` and a true double is reported as two simple eigenvalues.

Part 1: Tension curves on real corner domains

3 The roster and how the bases were built and validated

Everything measured before §20 used either the surrogate, a rectangle with plane waves, or the L-shape. The solver has to work on domains with *several* reentrant corners, where Fourier–Bessel families at those corners are what make it work at all (landscape_findings §20).

The first thing `curve_lab.py` had to get right is which corners need such a family. A corner is **singular** when its local orders $(\ell + \text{offset})\pi/\omega$ are not all integers, which under a single boundary condition is just "π/ω is not an integer". `Corner.is_singular` then meant "geometric corner",

a much larger and different set — all four corners of a rectangle were `is_singular` and none of them needs a Fourier–Bessel family, so the basis builder keyed on the fractional exponent itself (landscape_findings §20). That property is gone. The question is now asked where its answer actually lives, as `CornerModes.is_singular` (the offset comes from the boundary-condition pair through Grisvard's Robin rule, which is not a geometric fact); `geometry` keeps only the angle half of it, `Corner.is_reflection_closed`, meaning $\omega = \pi/m$. The builder's test is unchanged and is now one call: `CornerModes.from_corner(c).is_singular`.

Bases were built as plane waves sized by `n_useful(k*R)` plus 40 Fourier–Bessel terms per non-analytic corner. Crucially, each basis was *validated before* anything was concluded from its curves, by evaluating the tension at the tabulated eigenvalues (Table 1).

Table 1: The roster. Columns are basis size, corner structure, and the tension attained at the tabulated eigenvalues — the basis-adequacy check run before any curve was interpreted (landscape_findings §20).

domain	cols	corner structure	tension at table values
`L_shape`	134	1 reentrant	$10^{-8} \,..\, 2 \times 10^{-11}$
`mushroom`	168	2 reentrant	$7 \times 10^{-10} \,..\, 2 \times 10^{-10}$
`GWW1`	272	2 reentrant + two $3\pi/4$	$3 \times 10^{-9} \,..\, 10^{-9}$
`H_shape`	258	4 reentrant	$5 \times 10^{-8} \,..\, 2 \times 10^{-8}$
`plus`	248	4 reentrant, D_4	(no table)
`chevron`	252	an 18° spike	$2 \times 10^{-7} \,..\, 5 \times 10^{-8}$
`rect81`	176	no singular corner, control	(closed form)

`cut_square` was dropped from the roster. The `reference` recipe for it is an *origin-centred* Fourier–Bessel expansion of 320 terms, which the corner-adapted builder does not offer, and plane waves leave the tension at 10^{-2}. That is a basis-planning gap, not a curve finding (landscape_findings §20), and it is the reason the roster contains no curved boundary at all (see §19).

4 The plateau is stable across domains; the floor is not, within one scan

Table 2: Plateau height and the range of attained floors within a single scan (landscape_findings §20.1).

domain	plateau	floor min	floor max	spread
`L_shape`	0.1981	3.6×10^{-11}	9.1×10^{-7}	2.5×10^{4}
`mushroom`	0.1829	2.2×10^{-10}	1.4×10^{-8}	6.4×10^{1}
`GWW1`	0.1228	1.0×10^{-10}	2.6×10^{-8}	2.5×10^{2}
`H_shape`	0.2251	2.7×10^{-12}	1.2×10^{-7}	4.4×10^{4}
`plus`	0.2131	1.1×10^{-11}	3.7×10^{-8}	3.4×10^{3}
`chevron`	0.3555	4.9×10^{-8}	2.3×10^{-7}	4.7×10^{0}
`rect81`	0.1541	5.3×10^{-7}	4.9×10^{-2}	9.2×10^{4}

Two findings, and they point in opposite directions.

The plateau spans 0.12 to 0.36 across seven domains — a factor of 2.9, across areas from 1.0 to 14.0 and from zero to four reentrant corners. That is the strongest support yet for τ_{scan} as a *fraction of the plateau*, and against any absolute tension threshold (landscape_findings §20.1). Under the §2 rule, the plateau-relative form is admissible as a fixed constant and the absolute form is not.

The floor spreads four to five orders *within* a single scan on three of seven domains. The L-shape's five orders, reported earlier (search.md §1) and explained there by modes whose eigenfunction vanishes at the reentrant corner and is therefore analytic there, was not special. This is the direct justification for clause (2) of the operational definition being *per eigenvalue*: one `prec` target cannot be met uniformly when the attainable floor moves by 10^5 across the interval being scanned (landscape_findings §20.1).

`rect81` is the informative failure. Its floor maximum is 4.9×10^{-2}, which is no dip at all — about a third of its own plateau. A high aspect ratio with plane waves is a genuinely inadequate basis, and `rect81` is the only roster member whose audit flags for that reason rather than for multiplicity (landscape_findings §20.1). It recurs throughout Part 1 as the domain that no amount of sampling can fix.

5 Recall versus stride: six samples per gap is not enough

`search.md` §6 states that "roughly six samples per eigenvalue gap always place three inside every dip, and Weyl's law sizes that directly". That was *inferred* from the width/gap invariant on four benign domains. It was then measured end to end: subsample the fine scan at stride h and count how many true dips survive as discrete minima (Table 3).

Table 3: Recall as a function of stride, measured by subsampling the fine scans (landscape_findings §20.2). The first column is the coarsest stride, in samples per median gap, at which recall is still exactly 1.

domain	recall = 1.000 down to	at ~ 6 per gap	at ~ 4 per gap
`L_shape`	4.8 per gap	1.000	0.889
`mushroom`	5.0	1.000	0.917
`GWW1`	8.0	0.933	0.867
`H_shape`	9.1	0.875	1.000
`plus`	4.4	1.000	1.000
`chevron`	10.1	0.667	0.667
`rect81`	17.7	0.929	0.929

Full recall on every domain needs about 18 samples per median gap, not six. At six, four of seven domains lose eigenvalues (landscape_findings §20.2). This is a direct falsification of a standing design rule, and it is the single most consequential number in Part 1: the rule it replaces is cited in `search.md` §2.6 as part of the "not more work than asked" argument, and the replacement is three times as expensive.

Two qualifications are recorded in the source and are preserved here. First, 18 is a *fitted* number on seven domains, not a bound (solver_features §6). Second, the $\sim$ 4-per-gap column is not monotone with the $\sim$ 6 column on `H_shape` (1.000 against 0.875), which is a reminder that at coarse strides recall is a sampling accident rather than a smooth function of the stride.

5.1 The mechanism: minimum local gap, not Weyl's mean

The width/gap invariant is about the *mean* gap; recall is bounded by the *minimum local* gap. Measured, min(observed gap)/Weyl mean gap:

Table 4: Minimum observed gap as a fraction of the Weyl mean gap (landscape_findings §20.2).

domain	ratio	domain	ratio
`mushroom`	0.127	`chevron`	0.336
`H_shape`	0.210	`L_shape`	0.571
`rect81`	0.295	`plus`	0.740
		`GWW1`	0.792

The local gap runs 1.3× **to** 8× **tighter than Weyl's mean.** A stride computed from Weyl is therefore an estimate of the wrong quantity, and no a-priori stride can be safe — which is the case for correcting the stride *in the loop* rather than choosing it once (landscape_findings §20.2). This finding does more work than the 18-per-gap number itself: it is the reason Part 3's A2 (Weyl-stratified sampling) was tested at all, and the reason its falsification kills the whole a-priori-schedule family.

A third measurement in the same section corrects an earlier invariant. The half-width/gap ratio is *not* the near-constant reported in §14 of the notes. Measured consistently as the half-width at half the plateau, it runs 0.13 to 0.39 across the roster — a 3× range, with `H_shape` and `chevron` at the top. Section 14's 0.46–0.52 used a different definition (full width at the plateau) on four benign domains; converted, the roster runs about 0.5 to 1.6. **The invariant is much looser than one domain family suggested** (landscape_findings §20.2).

6 The Weyl audit is a post-multiplicity check

The Weyl deficit is the only completeness diagnostic available on a domain with no closed form: compare the number of levels found below a cut against the three-term asymptotic count $N(\lambda)$. When the audit was run at each stride, it flagged `MISS` at recall 1.000 on four of seven domains. That looks like a false positive and is not:

> σ_1 shows one dip per **level**, and Weyl counts **with multiplicity**. The deficit was reporting exactly the multiplicity that had not been estimated yet (landscape_findings §20.3).

Closing the loop — (C1) with $G = 10$ plus (C2) at prominence $P = 2$ on a wide bracket, then auditing the *unfolded* list — gives Table 5.

`plus` is the case that proves it: D_4 symmetry, two doubles found, and the deficit falls from 2.79 to 0.79. **(C1)+(C2) works on a domain with four reentrant corners and**

Table 5: Weyl deficit before and after multiplicity estimation (landscape_findings §20.3).

domain	dips	$\sum \mu$	deficit on dips	deficit after multiplicity
`L_shape`	9	10	0.41	0.41
`mushroom`	12	12	0.59	0.59
`GWW1`	15	15	0.33	0.33
`H_shape`	8	8	1.18	1.18
`plus`	8	10	2.79	**0.79**
`chevron`	3	3	0.12	0.12
`rect81`	14	14	2.55	2.55

a corner-adapted basis — the earlier 9-for-9 multiplicity score was on the disk and the equilateral triangle, both smooth (landscape_findings §20.3).

The residual flags are then real, and each names a different fault:

- `rect81` at 2.55 — an inadequate *basis*, floor 4.9×10^{-2} at some levels;
- `H_shape` at 1.18 — a genuine miss; its minimum observed gap is 0.378 against a Weyl mean of 1.795, so there is a tight pair the stride merged;
- `plus` at 0.79 — one more level or multiplicity not yet accounted for.

And `chevron` reads 0.12 at recall 0.667. **The audit is blind here**, because three distinct levels give too few cuts and every cut lies below the top level. That limitation is documented in `asymp.weyl_deficit`, and this is it firing on a real domain (landscape_findings §20.3).

6.1 What §20 changes

- τ_{scan} as a fraction of the plateau is now measured across seven domains, not assumed.
- The per-eigenvalue accuracy clause is necessary, not defensive: 10^5 floor spread in one scan.
- **The "six samples per gap" rule in `search.md` §6 is wrong and should be replaced.** About 18 per median gap reaches full recall on this roster, and even that is a fitted number rather than a bound, because it is the minimum local gap that governs.
- **The Weyl audit must run on the unfolded list, after multiplicity.** Run on dips, it reads as a false alarm on every symmetric domain — which is exactly the class the solver will meet at the optimizers of a shape optimization.
- The audit needs enough distinct levels to have cuts. Below about five it says nothing, and `chevron` shows it saying nothing while a third of the spectrum is missing (landscape_findings §20.4).

7 The Weyl deficit as a localizer, not a verdict

Since no a-priori stride is safe, the stride has to be corrected in the loop. `notes/adaptive_scan.py` tests one way: make the Weyl deficit a *profile* over cuts, so it says not only *that* a level is missing but *where*, and re-scan only that interval. Cost then goes as the number of tight spots rather than as the whole range (landscape_findings §21).

Scored against the §20 fine scans as truth, with $s_0 = 8$ (one sample per Weyl gap divided by 8), prominence $P = 2$, $G = 10$, $\tau_{\text{scan}} = 0.5\,S$:

Table 6: The audit-guided adaptive scan (landscape_findings §21). "found" includes spurious roots; the fine scan's evaluation count is given for comparison.

domain	truth	found	recall	spurious	deficit	evals	fine-scan n
`L_shape`	9	9	1.000	0	0.13	205	317
`mushroom`	12	11	0.917	0	0.47	238	348
`GWW1`	15	15	0.867	2	−0.09	326	483
`H_shape`	8	8	0.875	1	0.81	1556	376
`plus`	8	8	0.875	1	0.43	206	355
`chevron`	3	3	0.667	1	−0.78	84	154
`rect81`	14	17	1.000	0	0.94	4094	582

7.1 The bug it found first, which was the author's

The first version anchored the profile by subtracting $N(a)$, on the reasoning that the count is only meaningful relative to the bottom of the scanned range. **The three-term count is an asymptotic and goes negative at the low end** — measured $N(a)$ of -0.230 on `L_shape`, -0.687 on `H_shape`, -0.752 on `rect81`, -1.041 on `chevron`. Subtracting it *adds* that much to every deficit, so the loop chased a phantom:

Table 7: Effect of removing the spurious anchor (landscape_findings §21).

`plus`	1794 evaluations → 206, an 8.7× reduction, and the deficit reads 0.43 instead of 0.89
`L_shape`	deficit 0.36 → 0.13
`GWW1`	deficit −0.01 → −0.09

The count is absolute, so the correct form subtracts nothing and instead *requires* the found set to be a complete prefix from the bottom of the spectrum — the same precondition `asymp.weyl_deficit` already carries, and it should be asserted rather than papered over with an anchor that looks principled and is not (landscape_findings §21).

7.2 Two structural flaws, both real

(1) A cluster miss is *at* a level, not *between* levels. The localizer re-scans the interval between the two found levels straddling a hot cut. But when the missing eigenvalue is the second member of an unresolved pair, it sits at the *same* location as a found level, and no amount of re-scanning between levels will ever reach it. That is `mushroom` (recall 0.917, deficit 0.47, minimum observed gap 0.352 against a Weyl mean of 2.771) and `H_shape` (0.875, deficit 0.81, minimum gap 0.378 against 1.795). `H_shape` became *more* expensive after the anchor fix, 958 → 1556, because the corrected deficit is real and the loop keeps trying to fix it by the one mechanism that cannot. The deficit response therefore has to be two-pronged:

between levels	re-scan the interval	(a genuinely unsampled dip)
at a level	re-examine multiplicity	(an unresolved cluster)

and only the first is implemented (landscape_findings §21).

(2) No basis-adequacy termination guard. `rect81` spends 4094 evaluations — 7× the fine scan — and still reports a deficit of 0.94, because its floor reaches 4.9×10^{-2} at some levels and no stride can fix a basis. This is the unbounded-refinement stall, and the predicate that stops it already exists: gap $> 20\,\varepsilon/c$. Measured on `H_shape` at every level, gap$/(\varepsilon/c)$ runs 2.2×10^{6} to 1.4×10^{11} — massively adequate, so the guard correctly does *not* fire there and correctly separates "refine harder" from "the basis cannot do this". That discrimination is the thing the loop most needs and it is one predicate away (landscape_findings §21).

7.3 Two cautions on the scoring itself

`chevron` reads deficit -0.78 at recall 0.667: the audit says clean while a third of the spectrum is missing. Inside a control loop this is worse than in §6, because **the loop terminates on it**.

And the truth is basis-dependent. "Truth" here is the fine scan's dips below $0.5\,S$. On `rect81` the floor reaches 4.9×10^{-2}, about a third of the plateau, so real dips fall below that cutoff and the adaptive scan finding 17 against a "truth" of 14 is the truth undercounting, not the search over-reporting. **Any recall number on a domain whose basis is marginal is measuring the cutoff as much as the search** (landscape_findings §21).

Verdict. The localizer is worth keeping — it is what turned `plus` into 206 evaluations, and on the three domains where the miss is genuinely a resolution problem it reaches the fine scan's recall at 0.6× its cost. It is not sufficient on its own; the two additions it needs (the at-level prong, and the $20\,\varepsilon/c$ guard) are both specified and neither is speculative. What it should *not* be is a verdict (landscape_findings §21).

Part 2: What the solver has to have

8 Functional requirements

The list below is from `notes/solver_features.md`, derived from the operational definition of §2 and from the measurements of Part 1. Every entry names the measurement that forces it.

F1. Recall first, and recall is conditional.
Missing an eigenvalue cannot be repaired downstream; a located root a digit short can. Every claim of completeness is conditional on the scan's own assumptions and must say so.

F2. Adaptive subdivision, not a uniform grid.
A pair at separation d is resolved once sampling near it is finer than about $d/3$. A uniform grid pays that everywhere at $n \sim 3(b-a)/d$; subdivision pays $O(\log(1/d))$. Measured 2048 against 47 evaluations at $d = 3 \times 10^{-3}$, and 6×10^{10} against 97 at $d = 10^{-10}$. d is not known in advance, so the uniform cost is paid against a pair that may not exist. (search.md §3)

F3. A stride that is corrected in the loop.
Six samples per gap loses eigenvalues on four of seven real domains; full recall needs about 18 per median gap. And the governing quantity is the *minimum local* gap, which runs 0.13 to 0.79 of Weyl's mean — so no a-priori stride is safe and the correction has to happen during the solve. (landscape_findings §20.2), and §5 above.

F4. Per-solve calibration of c.
c runs 0.45 to 1.73 across domains, stable within one. A fixed Lipschitz constant is either wrong or wasteful. (landscape_findings §2)

F5. Threshold rank, re-derived at each λ.
The numerical rank *grows* with λ, so a rank frozen at one point is unsafe at the low end of a range: 39 columns chosen at $k = 3.95$ gives 85 spurious minima at $k = 2.35$, where the correct 31 gives exactly one. Threshold cannot pick an unsafe rank by construction, at 1.2–1.6× the cost. (landscape_findings §3, 5, 8)

F6. Two-level rank from *one* factorization.
The scan wants a low rank (smooth, wide dips); the refinement wants a high one (accurate floor). The L-shape locates λ_1 to 3.2×10^{-12} at rank 111 against 1.4×10^{-10} at rank 61, because what limits it is the floor rather than the roughness. One SVD serves both — no second basis, no second assembly. (landscape_findings §6) **Not built.**

F7. Refine in λ.
Fitting σ_1^2 over identical physical points, the residual is smaller in λ in 17 of 17 cases across four domains, median 5.1×, and the located vertex is 5.1× more accurate. The *scan* variable is a separate choice. (landscape_findings §3b)

F8. Squared objective.
Minimize σ_1^2, never σ_1: squaring removes the kink and makes refinement machine-exact. (search.md §3)

F9. Multiplicity by (C1) *and* (C2).
The ratio gap alone double-counts once $d \lesssim \varepsilon/c$, reporting multiplicity 2 at a *simple* eigenvalue; (C2) rejects it every time because σ_2 there is a monotone wall. Confirmed on a domain with four reentrant corners: on `plus_shape` the two D_4 doubles are found and the Weyl deficit falls $2.79 \to 0.79$. (landscape_findings §9.4, 20.3)

F10. Collapse-versus-merge by the interior Gram angle.
Two roots at the same separation can be one eigenvalue found twice or two genuine ones; no length threshold decides it. Distinct pairs read $|\cos| \leq 0.10$, duplicates ≥ 0.948. Within one λ the test is *vacuous* — the GSVD makes the interior images orthogonal by construction, measured 4.6×10^{-16} — so it is only informative *across* roots. (landscape_findings §15)

F11. Work scaled to `prec`.
Refinement stops at $\max(\texttt{prec} \cdot \lambda,\ \delta_{\text{res}})$; the first term binds whenever the basis is better than the request. Completeness is *not* a precision question and the stride must not tighten with `prec`. (search.md §2.6)

F12. Deterministic output.
The same domain returns the same eigenvalue twice. An unseeded interior fill gives a 110× run-to-run spread in the located eigenvalue and puts that into the shape objective as noise. (landscape_findings §17)

9 Required diagnostics

These are output, not instrumentation. S is unknown on every domain that matters, so they are the only signals that F1 has failed.

D1. The Weyl deficit, on the *unfolded* list.
The only completeness check that does not depend on the search being right. It found the recursion-floor bug that prefix truth did not. Two hard limits, both measured firing on real domains: run *before* multiplicity it reads as a false alarm on every symmetric domain (2.79 on `plus`, which is really 0.79), and below about five distinct levels it says nothing at all while a third of the spectrum is missing (`chevron`, 0.12 at recall 0.667). (landscape_findings §20.3)

D2. L and the largest observed secant slope, with their ratio.
A small margin means the constant was being pushed and the run deserves a rerun. The only signal that the first pass was too coarse. (See A6 in Part 3: this diagnostic is what would have caught the unsound pilot.)

D3. Per root: δ_{res}, and which termination bound was active.
This is what makes clause (2) enforceable while its constant is uncalibrated.

D4. The basis-adequacy predicate, per interval.
gap $> 20\,\varepsilon/c$, from a direct measurement — (C1) correct at $d/(\varepsilon/c) = 23$, wrong at 0.45 and 0.15. (landscape_findings §9.4)

D5. A flag when `tol_merge` approaches the local resolution limit.
At `prec` $\sim 4 \times 10^{-9}$ the discretization's own splitting of a degenerate pair exceeds `tol_merge` and a true double is reported as two simple eigenvalues. (landscape_findings §19)

D6. Evaluation count and attic revivals.

10 Anti-features: measured not to work

These carry their own numbering in `solver_features.md` (A1–A9) which is *unrelated* to the approach numbering A1–A27 of Part 3. They are relabelled X1–X9 here to avoid the collision.

X1. No absolute tension threshold, anywhere.
The floor spreads four to five orders *within* one scan on three of seven domains. Thresholds are fractions of the plateau, which is 0.12 to 0.36 across the whole roster. (landscape_findings §20.1)

X2. No σ_2-*value* gate for cluster detection.
σ_2's minimum is $c\,d/2$ only at the crossing point, which is never a sample, so a grid of spacing h sees $c\,h/2$. The gate tests the *grid*, not the pair. (landscape_findings §11)

X3. No offset detector expecting early warning.
The σ_2 offset relation is real (1% of $d/2$, confirmed over six decades) but it fires at exactly the same coarseness at which σ_1 first shows two minima. It measures d once resolved; it does not detect earlier. (landscape_findings §9.2, 16) (Part 3, §16, records the one detector

that *does* fire earlier, and the extent to which the corresponding general claim had been over-stated.)

X4. No distance-only collapse/merge.
See F10.

X5. No structured interior fill.
`method="grid"` is $30\times$ to $10^4\times$ more accurate as a discrete inner product *and* annihilates the mode side $+ 1$ exactly, to 10^{-56}, because its points sit on that mode's nodal lines. The interior block wants *non-degeneracy*, not accuracy. (landscape_findings §18)

X6. No interior-cubature project.
Accurate interior inner products for eigenfunctions are hard and the path is dysfunctional. The interior block only has to bound $\|u\|$ away from zero. Anything that must be *accurate* belongs on the Rellich boundary integrals.

X7. No fixed tolerance that expresses resolution or accuracy.
(search.md §2.1)

X8. No auto-swap to a symmetry-reduced solve on detected symmetry.
A tolerance threshold makes the objective discontinuous in shape, and it fires in the endgame of the optimization. (`mps_symmetry.md` §2.) This is the constraint that makes A25 in Part 3 a *declared*-symmetry option only.

X9. Do not design against a narrow dip.
No domain produces one; measured c is 0.445 to 1.725 over 19 eigenvalues. The real failure is a *cluster*. (landscape_findings §14)

11 The interface has to widen

The current harness contract is `search(tensions, a, b) -> [(lam, mult)]`, and `search.md` defends it: the search sees only the tension function, which is what makes candidates comparable and lets a later FEM or boundary-integral method reuse it.

F10 breaks that. The Gram angle needs the candidate eigen*functions*, which `tensions(x)` cannot supply. Two ways out (solver_features §4):

1. Widen the interface to `modes(x) -> (sigma[:], U[:, :])`, where U's columns are the candidate functions sampled on a *fixed* common point set. FEM and BIM produce eigenvectors too, so this stays method-agnostic and keeps candidates comparable. The fixed point set is already what `MPSPencil` has — `interior` is set at construction and only k changes.
2. Move the collapse phase above the search, into `MPSEigensolver`, as post-processing.

(1) is better, and it is a small change: the harness's `Case` grows one callable, and `uniform_grid_search` and the lappy baselines ignore it. (2) makes the collapse decision un-benchmarkable, which is precisely the decision that most needs benchmarking.

This finding is more consequential than it first looks, because Part 3's two surviving approaches — the eigenvector swing and the bisection repair — both read `modes(x)` and neither is expressible under the narrow contract.

12 Ranked build order

1. **The stride rule and the in-loop Weyl correction (F3, D1).** The measured defect is here — six per gap loses eigenvalues on four of seven domains — and recall is the headline.
2. **The interface widening (§11), then F10.** Everything about multiplicity is unbenchmarkable until the harness can see eigenfunctions.
3. **`prec`-derived termination (F11)** and the δ_{res} report (D3). This is what makes the precision contract real rather than aspirational; nothing currently reads `prec` at all.
4. **Two-level rank (F6).** A pure win already measured, and cheap: one factorization.
5. Basis planning, which bounds all of it and is a separate project.

Part 3: Candidate methods

13 The reframing that generates the list

This is not global optimization. Global optimization finds $\arg\min f$. What is needed is

$$\{\, x \in [a, b] \;:\; x \text{ is a local minimum of } \sigma_1 \text{ with prominence} \geq P \,\}$$

with a multiplicity at each, and a certificate that none was missed. That is a **level-set / sublevel-set problem** — find and certify $\{f \leq \tau\}$ — with a local-structure problem inside each component (approaches §0). Two consequences follow, and they organize everything below:

- the hard half is the **certificate** (what is provably absent), not the location;
- methods designed to converge fast to *one* minimum are mostly irrelevant, and methods designed to **count** or **enclose** are directly relevant even when they locate nothing.

The enumeration is therefore organized by *what supplies the certificate.*

14 The enumeration

14.1 Family 1 — deterministic sampling, local extraction

Certificate: "the samples were fine enough." The weakest kind, and everything currently built lives here.

A1. Uniform grid.
The control, `harness.uniform_grid_search`. Falsifiable claim: a grid at s samples per Weyl gap reaches recall 1.0 for $s \geq s^*$. Measured $s^* \approx 18$ on the roster, and it is not a bound (§5).

A2. Weyl-stratified grid.
Sample equispaced in $N(\lambda)$, not in λ. **Tested.**

A3. Recursive bisection into every dip.
`lappy_repaired_search`, 10/12 at 5573 evaluations. The open recursion-floor bug lives here.

A4. Audit-guided local refinement.
§7. Works where the miss is a resolution problem; structurally cannot reach a miss that sits *at* a level; no basis guard.

A5. Coarse-to-fine cascade sharing evaluations.
One sorted cache, repeated passes at halved stride, extraction from the cache each time.

14.2 Family 2 — bounding and certification

Certificate: a proven lower bound on f over an interval. The only family that can certify *absence* rather than assert it.

A6. Piyavskii–Shubert with a calibrated L.
The originating proposal. Built, broken, 7/12. **Tested.**

A7. DIRECT (dividing rectangles).
Lipschitz-free: no constant to calibrate, considers all Lipschitz constants at once by keeping the potentially-optimal set. **Tested.**

A8. Cone-model envelope.
Bound from the measured structure (1) rather than from secants: between two samples the cone model says how deep a dip can hide.

A9. Interval arithmetic / rigorous enclosure.
The only route to a true proof. Needs an interval extension of a generalized singular value through a GSVD — not available and not plausible to build. **Rejected**, with the reason recorded.

A10. Derivative-bounded envelope.
A two-point Hermite bound, far tighter than a secant bound. Gated on A14.

14.3 Family 3 — analytic root counting

Certificate: a contour integral counts the roots inside, exactly and with multiplicity. The only family that answers the completeness question *directly* rather than by sampling.

A11. Argument principle on $\det A_B(\lambda)$.
With a square collocation ($n_{\text{bdry}} = n_{\text{cols}}$), $\det A_B(\lambda)$ is analytic in λ and $(2\pi i)^{-1} \oint d(\log \det)$ counts its zeros. **Tested.**

A12. Beyn / Sakurai–Sugiura contour integration.
The standard tool for "all eigenvalues inside a contour, with multiplicity" for nonlinear eigenproblems $T(\lambda)v = 0$. Obstacle: MPS is rectangular and least-squares, so the moment matrices are not defined; a square reformulation exists only in A11's setting, which gives up least-squares stability.

A13. Inertia / Sylvester count.
$N(\lambda)$ exactly, by counting negative eigenvalues of a shifted Hermitian pencil. **Rejected**: FEM and BIM have this; MPS does not, because there is no Hermitian matrix whose inertia counts Laplacian eigenvalues below λ. It is recorded because it is the single feature whose absence makes this problem hard, and a hybrid borrowing a coarse FEM count purely as a counting oracle is a real option (A24).

14.4 Family 4 — derivative-based

A14. Exact $d\sigma/d\lambda$ from existing primitives.
Every particular solution here is $\psi(x) = \Psi(kx)$ about its own centre — plane waves $e^{ik\,d\cdot x}$ and Fourier–Bessel $J_\nu(kr)e^{i\nu t}$ both — so Euler's identity gives $\partial\psi/\partial k = (x \cdot \nabla\psi)/k$ exactly, and $x \cdot \nabla\psi$ is the *Rellich multiplier*, which the boundary-integral layer is already planned around. **Tested.**

A15. Sign-of-derivative bracketing.
A bracket is a sign change — no extra evaluations, and no reliance on three-point local-minimum tests that fail in the flat bottom.

A16. Curvature for δ_{res}.
The second derivative at the minimum gives δ_{res} directly rather than through a three-point parabola fit.

14.5 Family 5 — structure of the branch family

A17. Track all σ_j as smooth curves.
σ_1 is $\min_j$ of branches and has kinks at crossings; the individual branches are smooth. Continuation turns a non-smooth minimization into several smooth ones.

A18. σ_2 as a between-dip certificate.
Between two eigenvalues σ_1 is the meeting of two cones and σ_2 is bounded below there; a lower bound on σ_2 over an interval may certify "no third eigenvalue hiding here".

A19. Eigenvector continuation.
Track $c(\lambda)$ rather than $\sigma(\lambda)$. The coefficient vector swings as λ crosses an eigenvalue, and the swing is a large $O(1)$ signal where the tension change may be small. **Tested.**

14.6 Family 6 — model-based and statistical

A20. Gaussian process with a level-set acquisition.
Evaluations cost 14–100 ms, expensive enough that model-based placement can pay. Concern: a GP prior is stationary and σ_1 is emphatically not — deep narrow dips on a flat plateau. Would need the cone model as a mean function, which is A21.

A21. Parametric cone-mixture fit.
Fit $\sigma_1 \approx \min_i \sqrt{\varepsilon_i^2 + c^2(\lambda - \lambda_i)^2}$ over a window, then place the next sample where adding one more cone would be least detectable.

A22. Multistart local minimization with clustering.
No certificate at all, and the starts have to come from somewhere — which is A1. **Rejected as a primary**, kept as a refinement stage.

A23. Stochastic global search (annealing, evolutionary, CMA-ES).
Rejected. No completeness certificate, and clause (1) is a multiset match. These answer a different question.

14.7 Family 7 — exploiting problem structure

A24. A coarse FEM counting oracle.
Borrow $N(\lambda)$ from a cheap low-order FEM inertia count as the completeness certificate, keeping MPS for accuracy. It supplies precisely the missing A13, and the Weyl audit is a weak surrogate for it — Weyl is blind at the top of the range and needs five distinct levels; inertia is exact and local.

A25. Symmetry decomposition into irreps.
For a *declared* symmetry, each irrep has a simple spectrum and multiplicity becomes bookkeeping across irreps rather than a numerical test. Falsifiable on `plus_shape` (D_4), where the two measured doubles should split into distinct irreps. Constraint: declared only, never detected (X8).

A26. Two-level rank.
Low rank for the scan, high for refinement, one factorization. Already half-measured: rank 111 locates λ_1 to 3.2×10^{-12} against 1.4×10^{-10} at rank 61.

A27. Basis-adaptive escalation.
When the audit says the basis is binding (gap $< 20\,\varepsilon/c$), raise the basis rather than the sampling. Half-measured: `rect81`'s 4094 evaluations did not fix it (§7).

14.8 Triage

Table 8: Triage of the 27 candidates (approaches §8).

tier	approaches
TEST	A2, A6, A7, A11, A14, A19 — cheap, and each falsifies something load-bearing
KEEP	A3, A4, A5, A8, A15, A24, A25, A26, A27 — plausible, needs the tested ones first
LATER	A12, A17, A18, A20, A21 — real but expensive to evaluate
REJECT	A9, A13, A22, A23 — reasons recorded above

The six in TEST were chosen because each one, if it fails, removes a whole family: A2, if stratification does not beat uniform, no sampling schedule will and the answer must come from the certificate side; A6, if pilot secants underestimate the true slope, every Lipschitz certificate in the design is unsound as built; A7, if Lipschitz-free subdivision matches it, the calibration problem disappears; A11, if the phase of $\det A_B$ survives, we get an *exact* count, which nothing else offers; A14, if the derivative identity holds, four other approaches become available for one gradient; A19, if the eigenvector swing is coarser-samplable than the dip, the stride problem of §5 is solved outright (approaches §8).

15 The six tests

All six were run (`notes/test_approaches.py`). Table 9 is the summary; each is then given in full.

Table 9: Outcomes of the six TEST-tier approaches (landscape_findings §22).

	outcome	evidence
A2	FALSIFIED	worse or equal in 22 of 28 comparisons
A6	UNSOUND	`GWW1` at $n_{\text{pilot}} = 20$ reads 0.672: the certificate is *false*, not merely loose
A7	STRONG	mean recall 0.95 against uniform's 0.75, and no constant to calibrate
A11	FALSIFIED	455 spurious sign flips against 9 roots
A14	CONFIRMED	4.65×10^{-9} once the k-dependent column scale is corrected for
A19	REINTERPRETED	false as proposed; the peaks are crossings, not eigenvalues

15.1 A2 — Weyl-stratified sampling: falsified

What it is. Sample equispaced in $N(\lambda)$ rather than in λ, so that points concentrate where the spectral density is high.

The falsifiable claim. At equal point count, stratified recall $\geq$ uniform recall, with the gain largest where min gap/mean gap is smallest — i.e. on `mushroom` (0.127) and `H_shape` (0.210).

The evidence. Measured against uniform at equal point count, on all seven domains at $n = 40, 60, 90, 130$: **28 comparisons, and stratification is worse or equal in 22 of them.** At $n = 130$ it is worse on four of seven (landscape_findings §22).

Why it kills the family. The reason is structural. $N(\lambda)$ is nearly *linear* in λ, so stratifying by it is nearly uniform sampling, and the deviations it does introduce are not aligned with where the local gaps are tight. **Local clustering is a fluctuation phenomenon — level repulsion, accidental near-degeneracy — not a density phenomenon, and Weyl's law describes only the density.** No a-priori schedule computed from Weyl can concentrate samples where they are actually needed (landscape_findings §22), (approaches §9). This is the formal reason F3 must correct the stride *in the loop*: the information needed to choose it in advance does not exist in $N(\lambda)$.

15.2 A6 — Lipschitz calibration: unsound at small pilots

What it is. Piyavskii–Shubert branch-and-bound with $L = \gamma \max |d\sigma/d\lambda|$ estimated over n_{pilot} pilot points, $\gamma = 2$. This is the originating proposal for the whole search.

The falsifiable claim. The certificate is only as good as L, and L from pilot secants underestimates the true slope on some real domain.

The evidence. Ratios of the pilot-derived L to the true maximum slope from the fine scan; values above 1 mean the certificate holds (Table 10).

Table 10: A6: pilot-derived L against the true maximum slope (landscape_findings §22).

domain	true max	$n = 20$	$n = 40$	$n = 80$	$n = 160$
`L_shape`	0.2013	1.894	1.989	1.996	2.000
`mushroom`	0.3589	1.524	1.945	1.974	1.995
`GWW1`	0.7612	**0.672**	1.974	1.988	1.999
`H_shape`	0.2438	1.886	1.941	1.985	1.991
`plus`	0.2738	1.939	1.986	1.996	1.999
`chevron`	0.0472	1.993	2.000	2.000	2.000
`rect81`	0.2264	1.917	1.945	1.951	1.961

`GWW1` **at** $n = 20$ **reads** 0.672**: the pilot underestimates the true slope even with** $\gamma = 2$**, so the certificate is *false*.** It is the domain with the densest spectrum, and the pilot simply never lands on a steep wall (landscape_findings §22).

Why this is *unsound*, not merely loose. The distinction matters and is worth stating plainly. A loose constant costs evaluations; an *underestimated* one voids the guarantee. `search.md` §6's conditional claim — "no x with $\sigma_1(x) \leq \tau_{\text{scan}}$ outside the returned set, *given* L" — is conditional on something that can be false, and the failure mode is a silently missed eigenvalue rather than wasted work (approaches §9). Everything converges to 2.000 by $n = 40$, so the failure is not asymptotic; it is that "a fine enough pilot" is domain-dependent and not knowable in advance, which is §5's stride problem wearing a different hat. A6 is usable only if something else sizes the pilot — which is why it survives as WOUNDED rather than dead.

15.3 A7 — DIRECT, Lipschitz-free: strong

What it is. Keep every interval that is potentially optimal for *some* Lipschitz constant — the lower-right convex hull of (width, value) — plus every interval already below τ. There is no constant to calibrate, so A6's failure mode cannot occur.

The falsifiable claim. Extended to keep every interval potentially containing a point below τ, DIRECT reaches recall 1.0 at fewer evaluations than a uniform grid.

The evidence. At about 230 evaluations (Table 11).

Six of seven at or above uniform, mean recall 0.95 **against** 0.75, and it beats §7's audit-guided loop on `mushroom`, `chevron` and `plus` while costing 7× less than it on `H_shape` and 18× less on `rect81` (landscape_findings §22).

Caveat, recorded. This simulation interpolates from the cached fine scan, so it does not pay real evaluation cost and the interpolation smooths the curve. The structural result — no constant needed, and it beats uniform — does not depend on that (landscape_findings §22).

Table 11: A7: DIRECT against a uniform grid at the same evaluation count (landscape_findings §22).

domain	DIRECT recall	uniform recall at the same n
`L_shape`	1.000	0.778
`mushroom`	1.000	0.917
`GWW1`	0.800	0.867
`H_shape`	0.875	0.625
`plus`	1.000	0.875
`chevron`	1.000	0.333
`rect81`	1.000	0.857

15.4 A11 — the argument principle on $\det A_B$: falsified

What it is. With a square collocation, $\det A_B(\lambda)$ is analytic in λ and its sign changes count the odd-multiplicity roots. This was the only exact-count candidate available to MPS.

The falsifiable claim, and why it was not the usual objection. The determinant method is known to be numerically bad — that is exactly why Betcke and Trefethen introduced the subspace angle and why this project uses the GSVD. But that objection is about the *magnitude*, and counting needs only the *sign*, which comes from LU pivot parity and is a different question. The narrow claim was that the phase might survive where the magnitude does not (approaches §3).

The evidence. It is not a different question. On 900 samples:

Table 12: A11: sign flips of $\det A_B$ against the true eigenvalue count (landscape_findings §22).

domain	$\log\lvert\det\rvert$ span	sign flips	true eigenvalues
`L_shape`	214 orders	464 (455 spurious)	9
`mushroom`	163 orders	456 (444 spurious)	12

Every true eigenvalue does have a flip near it — 9/9 and 12/12 — but at 2% precision that is worthless.

The finding, which is stronger than the standard objection. The conditioning destroys the determinant's *topology*, not merely its magnitude (landscape_findings §22). That is a strictly stronger statement than the usual complaint about determinant methods, and it kills A11 and most of A12 (Beyn on a square MPS collocation) with it, because they rest on the same matrix. With A13 already rejected, MPS is left with no exact counting oracle of its own, which is what makes A24 (borrowing one from FEM) the standing option.

15.5 A14 — the exact k-derivative: confirmed, after a correction

What it is. Every particular solution here is $\Psi(kx)$ about its own centre, so Euler's identity for a homogeneous argument gives

$$\frac{\partial \psi}{\partial k} \;=\; \frac{x \cdot \nabla \psi}{k}$$

exactly. $d\sigma_1/d\lambda$ then follows by first-order perturbation of the GSVD.

The falsifiable claim. The analytic derivative matches a central finite difference to the accuracy the finite difference deserves.

The evidence, including the initial failure. Measured against a central difference:

`PlaneWave`	3.12×10^{-10}	(finite-difference noise floor 2.23×10^{-10})
`FB @ corner`	2.35×10^{1}	$\rightarrow$ 4.65×10^{-9} after the correction below

The raw identity fails for Fourier–Bessel, and the reason is the k-dependent *column scale*. With $\psi_s = e^{-L(k)}\psi$ and L spatially constant,

$$\frac{\partial \psi_s}{\partial k} \;=\; \frac{x \cdot \nabla \psi_s}{k} \;-\; L'(k)\,\psi_s,$$

and $|dL/dk|$ reaches 1.74 for the corner family against exactly 0 for plane waves — which is precisely why one passed and one did not. With the correction both sit at the finite-difference floor (landscape_findings §22). The episode is worth keeping in the record: the identity is exact mathematics, and what broke it was an implementation detail of the equilibration, not the mathematics.

What it opens. $d\sigma/d\lambda$ is available for one gradient evaluation plus a scalar per column, rather than two extra tension evaluations. And $x \cdot \nabla \psi$ is the Rellich multiplier, which the boundary-integral layer is already planned around. This opens A10 (a Hermite envelope, far tighter than a secant bound), A15 (sign-change bracketing with no extra evaluations), A16 (curvature for δ_{res} directly), and a c calibrated exactly rather than by secants — which is the direct repair for A6 (landscape_findings §22), (approaches §9).

15.6 A19 — the eigenvector swing: falsified as proposed, strong as reinterpreted

What it is, as proposed. Track $c(\lambda)$ rather than $\sigma(\lambda)$: the angle between $c(\lambda)$ and $c(\lambda + h)$ was hypothesized to peak *at* eigenvalues, giving an indicator that survives a coarser stride than σ_1's dip.

The evidence against the original hypothesis. It does not: recall 0.333 **at 4 samples per gap, falling to** 0.000 **at 8 and finer** (landscape_findings §22). The finer the stride, the worse it gets — the signature of an indicator that is firing somewhere other than where it was expected.

The reinterpretation. The swing peaks where the *minimising vector changes identity*, which is at a branch *crossing* — and crossings lie *between* eigenvalues. This is far more useful than the original hypothesis: a peak in every gap partitions $[a, b]$ into cells holding exactly one eigenvalue each, which is a completeness certificate of a kind no sampling schedule gives. Measured as "gaps hit" out of the gaps between known dips (Table 13).

Table 13: A19b: the swing as a cell decomposition (landscape_findings §22). Timing is per sample, for `coef`.

domain	3/gap	4/gap	6/gap	8/gap	extra peaks at 4/gap	ms per sample
`L_shape`	8/8	8/8	8/8	9 (1 extra)	0	13.7
`mushroom`	10/11	11/11	11/11	11/11	0	32.7
`GWW1`	14/14	14/14	14/14	14/14	0	111.3
`H_shape`	7/7	7/7	7/7	7/7	0	81.7
`plus`	7/7	7/7	7/7	7/7	0	78.9
`chevron`	2/2	2/2	2/2	2/2	0	106.9
`rect81`	12/13	12/13	13/13	13/13	0	10.7

At four samples per Weyl gap, six of seven domains give a perfect cell decomposition, and `rect81` needs six. Against the 18 per median gap that σ_1 dip detection needs for full recall (§5), that is a 3× to 4.5× reduction in samples.

And it is nearly free. `coef` costs 13.7 ms against `tensions`' 13.6 on `L_shape`, 111.3 against 104.8 on `GWW1`, 81.7 against 76.6 on `H_shape` — **5 to 7 percent** — because both need the same regularization and factorization, and the `separate`-mode GSVD is comparable to `gsvdvals` (landscape_findings §22).

Extra peaks are 0 or 1–2 and are the safe direction: an extra cell is an empty cell, which is cheap to detect and discard, whereas a missing cell is a missing eigenvalue.

Three caveats, all real and all recorded. The denominator is gaps between *detected* dips, so this measures "does it separate the levels we already know about", not "does it find a hidden pair" — the pair-resolution question is untouched. The 5 × median peak threshold is a parameter and has not had the margin study that G and τ_{gram} got. And it *brackets* rather than locates: refinement still runs inside each cell, though refinement inside a bracket known to hold exactly one minimum is the easy and robust case, which is the entire point (landscape_findings §22).

15.7 Shortlist after the six tests

KEEP, TESTED	A7 (DIRECT), A19b (swing cell decomposition), A14 (exact derivative)
KEEP, UNTESTED	A3, A5, A8, A15, A24, A25, A26, A27
DEAD	A2, A11, A12-square, A9, A13, A22, A23
WOUNDED	A6 — unsound at small pilots; usable only with A7 or A19b sizing it

The two survivors attack the same defect from opposite sides. **A19b gives the partition; A7 gives the search inside it, with no constant to calibrate.** They compose: use the

swing at 4 per gap to cut $[a, b]$ into one-eigenvalue cells, then run DIRECT inside each cell against τ, then refine. That combination has a completeness argument that neither the Lipschitz certificate (A6, measured unsound) nor the Weyl audit (§6, blind at the top and needing five levels) can provide, and it is the first candidate in this project that does (landscape_findings §22).

16 The swing on a pair: a detection result

§15.6 left A19b's central question open: its cells were measured against gaps between *detected* dips, so it had never been shown to split a pair that σ_1 presents as one dip. That is the failure mode that (landscape_findings §9.4) and (landscape_findings §19) call real, and it decides whether the composed method is a capability win or only a cost win. The instrument is `rect(1+delta, 1)` with modes $(2, 1)$ and $(1, 2)$, in one window holding the pair and no neighbour (`notes/swing_pairs.py`).

Table 14: Pair detection at stride $h = 1.005$, about 12 samples per gap (landscape_findings §23).

$d(\lambda)$	σ_1 dips	swing peaks	verdict
5.139×10^{0}	2	1	σ_1 already splits
1.700×10^{0}	2	1	σ_1 already splits
5.834×10^{-1}	1	1	swing sees it, dip does not
1.769×10^{-1}	1	1	swing sees it
5.913×10^{-2}	1	1	swing sees it
1.776×10^{-2}	1	1	swing sees it
5.921×10^{-3}	1	1	swing sees it
1.776×10^{-3}	1	0	neither
0 (exact)	1	0	degenerate control

At that stride σ_1 stops splitting the pair below $d \sim 1.7$, and the swing keeps splitting it down to $d \sim 5.9 \times 10^{-3}$: **a factor of about 290**, at the same sampling cost. In stride units the swing fires at $d/h = 5.9 \times 10^{-3}$ — it resolves a pair about 170× finer than the sample spacing (landscape_findings §23).

This contradicts a recorded conclusion. `search.md` §3 says "there is no detector here that sees a pair before σ_1 does", and (landscape_findings §11) measured the offset detector firing "at exactly the same coarseness at which σ_1 first shows two minima". Both are correct *about the offset*, and both were generalized into a claim about detectors in general. The swing is a counterexample, and it works because it keys on something neither of those does: **the identity of the minimising eigenfunction, not the shape of the tension curve.** The two eigenfunctions stay near-orthogonal at $|\cos| \sim 1/\sqrt{n_{\text{int}}}$ far past where the tension merges them, and that orthogonality is what survives (landscape_findings §23).

The degenerate control is the other half of the result. At $\delta = 0$ the swing reads *zero* peaks. That is correct and not a failure: a true double has no crossing, because there is no other branch for the minimiser to switch to. So

swing peak inside a dip	→ two *distinct* eigenvalues
no swing peak	→ a genuine degeneracy, *or* a pair below the swing's own limit

The first arrow is a multiplicity discriminator of a kind (C1)+(C2) does not have: the double-count case (the gap rule reporting multiplicity 2 at a *simple* eigenvalue once $d \lesssim \varepsilon/c$) is exactly where the swing says "distinct", whereas (C2) currently has to reject it by a monotonicity argument on σ_2. The second arrow is still ambiguous, so this does not remove the resolution limit; **it moves it** (landscape_findings §23).

Where the limit sits, and it is not the pencil.

$h = 3.770$	fires to $d = 5.9 \times 10^{-2}$, not 1.8×10^{-2}
$h = 1.885$	fires to $d = 5.9 \times 10^{-2}$, not 1.8×10^{-2}
$h = 1.005$	fires to $d = 5.9 \times 10^{-3}$, not 1.8×10^{-3}

$d_{\min}$ tracks h, roughly $h/30$ to $h/170$. It is *not* ε/c, which is about 3.5×10^{-9} here — six orders below where the swing stops. **So the swing's limit is set by the stride and the peak threshold, not by the basis**, and there is a great deal of headroom left: a finer stride buys proportionally finer pair detection, which is not true of any dip-based detector (landscape_findings §23).

Caveats. One domain, one mode pair, one basis, thresh = 5 × median. The threshold has had no margin study — G and τ_{gram} both turned out non-delicate and this one has not been checked at all. And $d_{\min} \sim h/30$ to $h/170$ is a 5× spread across three strides, so the scaling is indicative rather than established (landscape_findings §23).

17 The composed candidate, scored

`tests/searches.py::swing_direct_search`: swing cells at 6 samples per estimated gap, DIRECT inside each cell against $\tau = 0.5\,S$, Brent in the resulting one-root bracket, (C1)+(C2) for multiplicity. Scored through the harness on five real domains with `reference` table truth, windows cut inside a tabulated gap so the truth is *complete*:

`L_shape`	$[4.82, 49.35]$	9	`GWW1`	$[1.27, 11.94]$	8
`mushroom`	$[2.75, 33.39]$	8	`H_shape`	$[3.87, 25.58]$	8
`chevron`	$[19.83, 192.25]$	8			

Table 15: End-to-end scores on the five-domain roster (landscape_findings §24).

method	mean recall	total evals	spurious	mult errors
`uniform_grid`	0.881	1010	0	0
`lappy_repaired`	0.319	1946	0	0
swing + DIRECT	0.975	2221	0	0

Table 16: Per case, swing + DIRECT (landscape_findings §24). The exclamation mark on `H_shape` is the Weyl deficit exceeding its 0.5 threshold — the flag firing on the one miss.

case	recall	prec	miss	spur	multX	max err	evals	weyl
`L_shape`	1.000	1.000	0	0	0	2.3×10^{-8}	473	0.41
`mushroom`	1.000	1.000	0	0	0	4.1×10^{-9}	442	0.02
`GWW1`	1.000	1.000	0	0	0	8.8×10^{-9}	454	0.33
`H_shape`	0.875	1.000	1	0	0	8.0×10^{-9}	385	0.76 !
`chevron`	1.000	1.000	0	0	0	9.2×10^{-9}	467	0.12

Four of five perfect, no spurious roots and no multiplicity errors anywhere, and the one miss is *flagged* — `H_shape` reads 0.76 against the 0.5 threshold. That is clause (3) of the operational definition doing its job: the failure is reported rather than silent, which is the difference between recoverable and fatal inside a shape-optimization loop (landscape_findings §24).

17.1 The honest ledger

It costs 2.2× **the uniform grid** for +0.094 recall. Recall is the headline metric by design — a missed eigenvalue cannot be repaired downstream — so this is the right trade under the stated principle, but **it is a trade and not a free win** (landscape_findings §24).

`lappy_repaired`'s 0.319 **is not a fair comparison.** It is starved by the harness's crude `_spacing` estimator, exactly the caveat (landscape_findings §11) records: a coarse pass sees a pair as one dip, concludes the interval holds one eigenvalue, and sizes a 20-point grid. **The number measures the estimator, not lappy's architecture.** Its own recall table, with `n_pts` set explicitly, is 10/12 (landscape_findings §24).

The accuracy column is F11 working, not a defect. swing+DIRECT reports maximum errors of 4×10^{-9} to 2×10^{-8} against a requested `prec` $= 10^{-8}$; the uniform grid reports 8×10^{-10} to 6×10^{-9} because it refines to `xatol` $= 10^{-13}$ regardless of what was asked. The composed search stops at $\max(\texttt{prec} \cdot \lambda,\ \delta_{\text{res}})$ and the first term binds. Doing better than asked is allowed; doing it at 2× the refinement cost on every root is what (search.md §2.6) rules out (landscape_findings §24).

17.2 Two efficiency findings

The first version re-scanned the *cell* grid to estimate the plateau, having already scanned a 64-point pilot for the spacing. The plateau is a median over an $O(1)$-varying curve and needs no cell-stride resolution; folding both into one 48-point pilot cut 2535 evaluations to 2221 with identical recall.

Dropping to 4 samples per gap and a DIRECT budget of 10 cuts cost further to 1583 — 1.6× — but recall falls to 0.925 (`H_shape` 0.875 → 0.750, `chevron` 1.000 → 0.875). §15.6 measured 4 per gap as sufficient for the *cells*, and it is; **what fails at 4 is the DIRECT search inside them**. The two budgets are separable and should be tuned separately (landscape_findings §24).

17.3 What the roster scoring left untested

Five domains, all with table truth, all at low λ. No pair below the swing's own limit appears in any of these windows, so §16's 290× is not exercised here — the composed search had not yet been scored on a case where it should beat a dip-based method outright. `rect(1+delta, 1)` is the instrument and it was not wired into the harness (landscape_findings §24). That is what §18 does, and what it found was a correction.

18 Detection is not separation: a correction to the pair result

`rect(1+delta, 1)` was wired in as harness cases: window $[10, 88]$, holding $(1, 1)$, the $(2, 1)/(1, 2)$ pair and $(2, 2)$; truth from the closed form and *complete*; δ swept so the pair separation runs 5.8×10^{-1} down to 0.

The first run failed, at every separation including the widest. All three methods scored recall 0.750 — three of four — and the degenerate $\delta = 0$ case scored 1.000. A window whose *split* cases fail while its *degenerate* case passes says the pair is being merged, not that a level is missing (landscape_findings §25).

18.1 The defect

Instrumenting the $\delta = 10^{-2}$ case, $d = 5.83 \times 10^{-1}$:

pilot sees 3 wells in a 4-eigenvalue window	→ estimated gap 26.0
cell grid $n = 19$ over $[10, 88]$	→ stride $h = 4.33$
crossing truly at 48.86	
swing peak *reported* at 46.83	
cells $[10,\ 29.5,\ 46.83,\ 64.17,\ 88]$	→ **both pair members in one cell**

The swing *detected* the crossing. It localized it only to $\pm h/2$, and with $d < h$ the cell boundary landed outside the pair instead of between its members.

The self-correction, stated plainly. Section 16 measured detection, not separation, and one was generalized into the other. Its in-pair test counted a peak as in-pair if it lay within $[\lambda_a - h,\ \lambda_b + h]$ — a tolerance of a full stride on each side. Every number in §16 stands as a statement about *detection*; none of them was ever a statement about splitting. **A pair detector and a pair separator are different things, and the 290× is the first, not the second** (landscape_findings §25).

18.2 The repair: a peak is a feature to bisect toward

The crossing lies in whichever half of an interval carries the larger angle; the halves are equal width so the comparison is fair; and each level costs one mode evaluation. Cost is $\log_2(h/d)$ — the same logarithmic behaviour adaptive subdivision has on the tension, obtained here from the eigenvector instead — and it converges to the crossing *regardless of how coarse the initial stride was*. That last property matters, because the stride here was set by a pilot that merged the pair and therefore could not have been chosen correctly in advance (landscape_findings §25).

Table 17: swing+DIRECT on `rect(1+delta, 1)`, before and after crossing localization (landscape_findings §25).

$d(\lambda)$	recall before	recall after	`uniform_grid`	`lappy_repaired`
5.834×10^{-1}	0.750	1.000	0.750	0.250
1.769×10^{-1}	0.750	1.000	0.750	0.750
5.913×10^{-2}	0.750	1.000	0.750	0.250
1.776×10^{-2}	0.750	1.000	0.750	0.250
5.921×10^{-3}	0.750	0.750	0.750	0.750
1.776×10^{-3}	0.750	0.750	0.750	0.500
5.922×10^{-4}	0.750	0.750	0.750	0.500
0 (exact double)	1.000	1.000	1.000	0.250
mean	0.781	0.906	0.781	0.438

The uniform grid never splits this pair at any separation tested, including the widest, because its stride comes from the same merged-well estimate. The composed search splits it down to $d = 1.78 \times 10^{-2}$, so its advantage over the control is at least $32\times$ in separation, and the true factor is larger since the control's own limit is above 5.8×10^{-1}.

The roster of §17 is unchanged at mean recall 0.975, at 2596 evaluations against 2221 — localization costs about 17% and buys the pair regime (landscape_findings §25).

18.3 The remaining frontier

At $d \leq 5.9 \times 10^{-3}$ the composed search still scores 0.750 with one spurious root and one multiplicity error. The cell boundary *is* found (§16 says the swing detects to 5.9×10^{-3}), but the two refinements in adjacent cells converge to the same minimum. The tension floor at this pair is 9.1×10^{-8} to 1.1×10^{-7}, so ε/c is about 2×10^{-6} and $d = 5.9 \times 10^{-3}$ is $3000\times$ above it: **the pair is resolvable in principle and this is a search failure, not a basis limit.** Two candidates, neither tested (landscape_findings §25):

1. DIRECT's fixed budget is too small for a cell whose dip sits near its edge;
2. the Gram-angle collapse phase is *not implemented* in `swing_direct_search`, so a duplicate arising from two adjacent cells is reported rather than collapsed and recounted.

The second is a direct instance of the interface finding of §11: the collapse decision is exactly the one the harness could not see before `modes(x)` was available.

19 Open questions and what is untested

Collected from the "what was not checked", "still untested" and "remaining frontier" passages of the sources.

On the stride rule. The 18-per-gap number is a *fitted* value on seven domains, not a bound. The governing quantity — the minimum local gap — is not predictable a priori, so this can only ever be a starting point for F3's in-loop correction (solver_features §6).

On the accuracy contract. C in the accuracy clause remains uncalibrated; clause (3) is the workaround. And the wide bracket used for (C2) was taken as 0.02λ, an ad-hoc fraction; it should be derived from the refinement's own parabola, and is not (solver_features §6).

On the roster's coverage. Everything above is at low λ — the roster tops out at 90 and mostly sits at 30–40. There is no domain with a hole, and no curved boundary except `cut_square`, which was dropped for a basis-planning reason rather than a curve one (solver_features §6), (landscape_findings §20). The composed candidate's five-domain scoring is likewise all low λ and all table truth (landscape_findings §24).

On the swing. One domain, one mode pair, one basis for the pair study. The $5\times$ median peak threshold has had no margin study, unlike G and τ_{gram}. The $d_{\min} \sim h/30$ to $h/170$ scaling is indicative, not established — a $5\times$ spread over three strides (landscape_findings §23).

On DIRECT. The A7 numbers come from a simulation that interpolates the cached fine scan, so it pays no real evaluation cost and the interpolation smooths the curve (landscape_findings §22).

On the sub-5.9×10^{-3} pair regime. Both candidate explanations — an under-sized DIRECT budget in an edge-dip cell, and the missing Gram-angle collapse — are untested (landscape_findings §25).

Structurally unbuilt. F6 (two-level rank from one factorization) is measured and not built. The at-level prong of the deficit response and the $20\,\varepsilon/c$ termination guard (§7) are both specified and neither is implemented. The interface widening of §11 gates everything about multiplicity.

Still open in the enumeration. A24 (a coarse FEM inertia count as a counting oracle) is the standing answer to the fact that A11 and A13 are both dead and MPS therefore has no exact count of its own. A25 (declared-symmetry irrep decomposition) would remove the (C1)/(C2) test entirely for symmetric domains — the class the solver will meet at the optimizers of a shape optimization — and is untested. A8 (the cone-model envelope), A15 (sign bracketing, now available since A14 is confirmed), A26 and A27 are all in the KEEP, UNTESTED tier.

Note on two inconsistencies in the sources

Two points were left as the sources have them rather than silently resolved.

First, `docs/design/search.md` §2.6 still states the scan stride as "about six samples per eigenvalue gap" as part of its "not more work than asked" argument, while `landscape_findings` §20.4 explicitly says that rule "is wrong and should be replaced". The design document has not been updated. The clause it is attached to — completeness is not a precision question, and the stride does not tighten with `prec` — is unaffected by the correction.

Second, the anti-feature list in `solver_features.md` §3 uses the labels A1–A9 for a set of things measured not to work, while `approaches.md` uses A1–A27 for the candidate methods. The two numberings are unrelated and collide (for instance, anti-feature A2 forbids a σ_2-value

gate, while approach A2 is Weyl-stratified sampling). The anti-features were relabelled X1–X9 in §10 of this report; the underlying notes should probably adopt a distinct prefix.